ONLINE RETAIL

BUSINESS INTELLIGENCE

Dec 2010 _____ Dec 2011

Sales Performance, Customer Behavior & Business Risk Analysis

Prepared by **Dihan** — Data Analyst

EXECUTIVE SUMMARY

Business Highlights & Key Focus

- 84.59% of sales revenue came from the UK.
- 65.57% of identifiable customers were repeat customers.
- The top 10% of identifiable customers generated 61.41% of identifiable customer revenue.
- 337 high-value customers were classified as at-risk.

Orders - 20,726

Sales Revenue - £10.64M

Identifiable Customers - 4,339

Average Order Value - £513.47

BUSINESS & DATA OVERVIEW

About The Business & Data

UK-based online retailer with international customers, analyzing sales, customer behavior, product performance and commercial risk.

Dec 2010 - Dec 2011

Dataset facts:

- 541,909 raw records
- 536,641 cleaned records
- 38 countries
- 4,339 identifiable sales customers
- Dec 2010–Dec 2011
- UCI Online Retail dataset

Tools

· Python
· Pandas
· SQL
· Power BI

WorkFlow

RAW DATA →

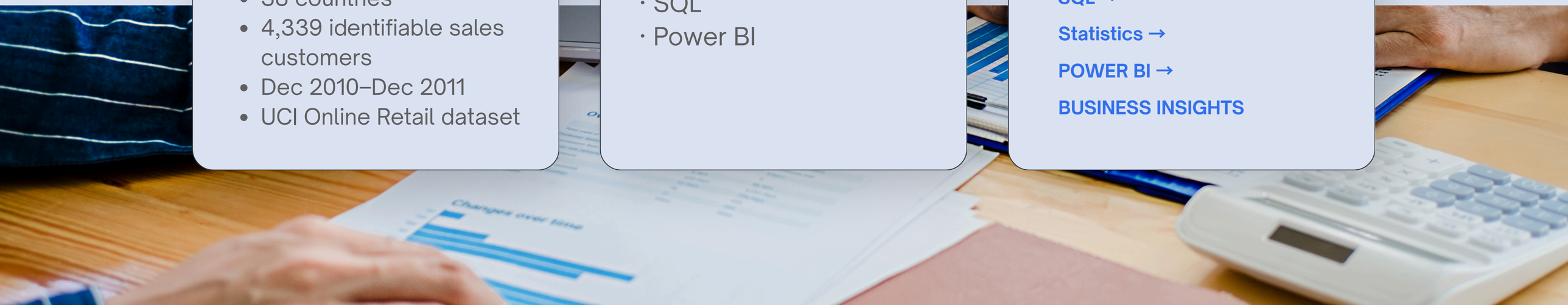
CLEANING →

SQL →

Statistics →

POWER BI →

BUSINESS INSIGHTS



SALES PERFORMANCE

Monthly Revenue Trend

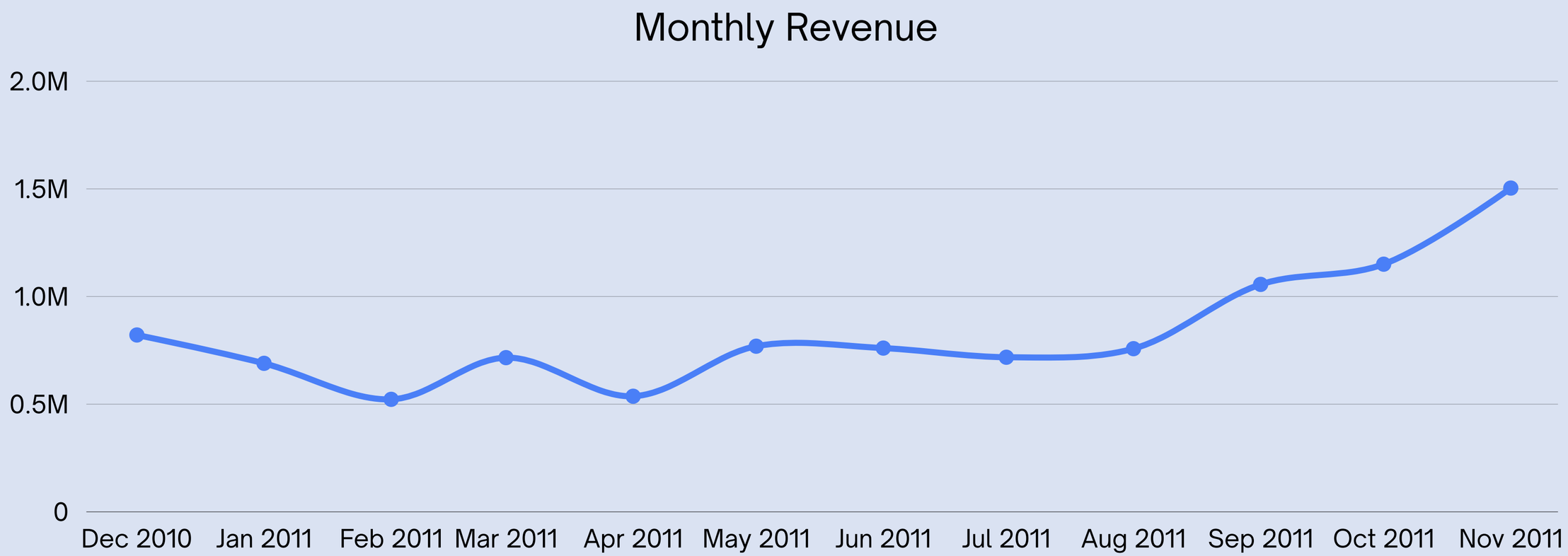
£10.64M
Total sales

£513.47
Averageorder value

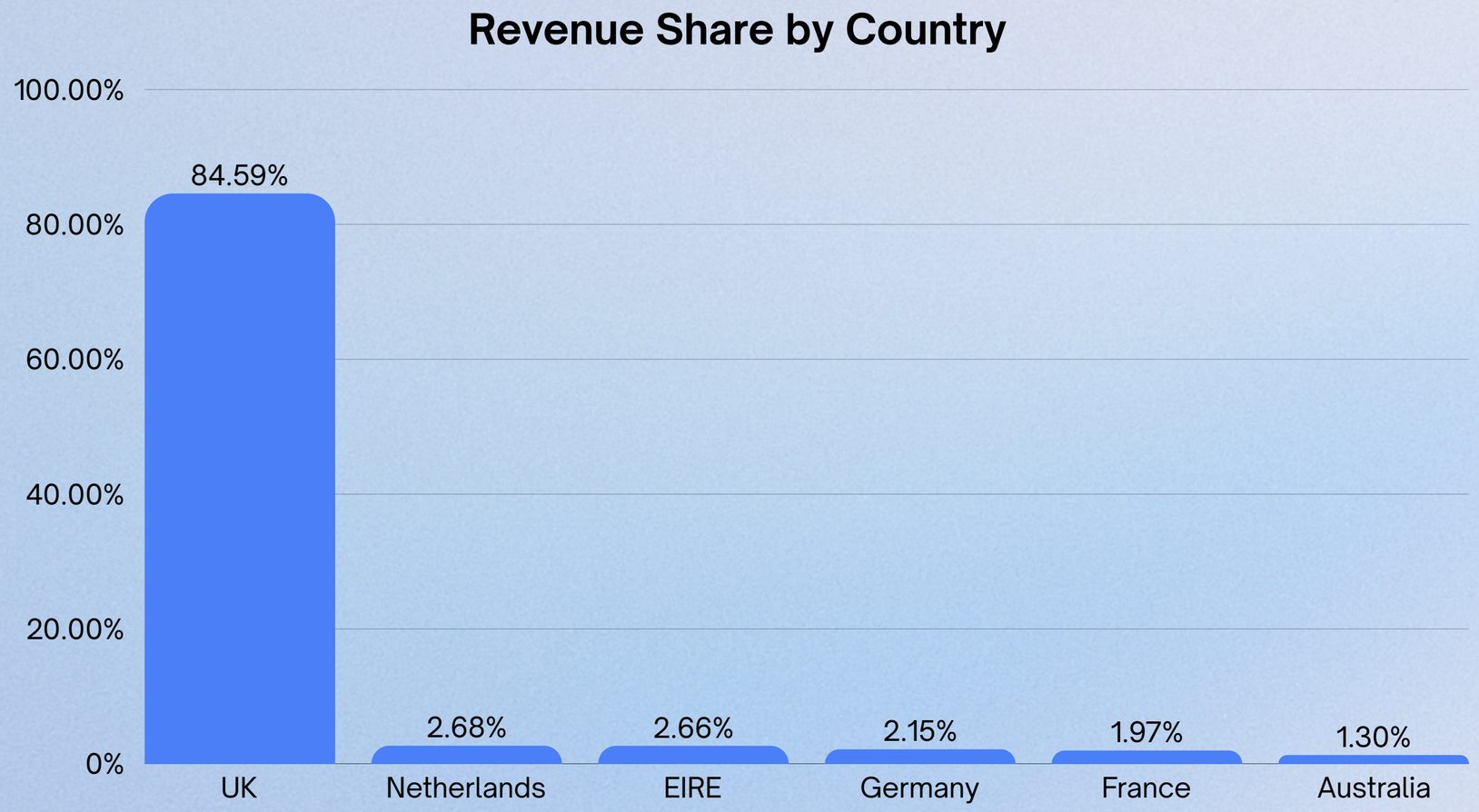
£1.50M
November 2011 revenue

Growth Rate
+ 83% Dec 2010 → Nov 2011

Sales increased substantially during the second half of 2011, with November representing the highest complete month in the dataset.



GEOGRAPHIC PERFORMANCE



UK Revenue Share

+84% Primary market dependency

Next major markets

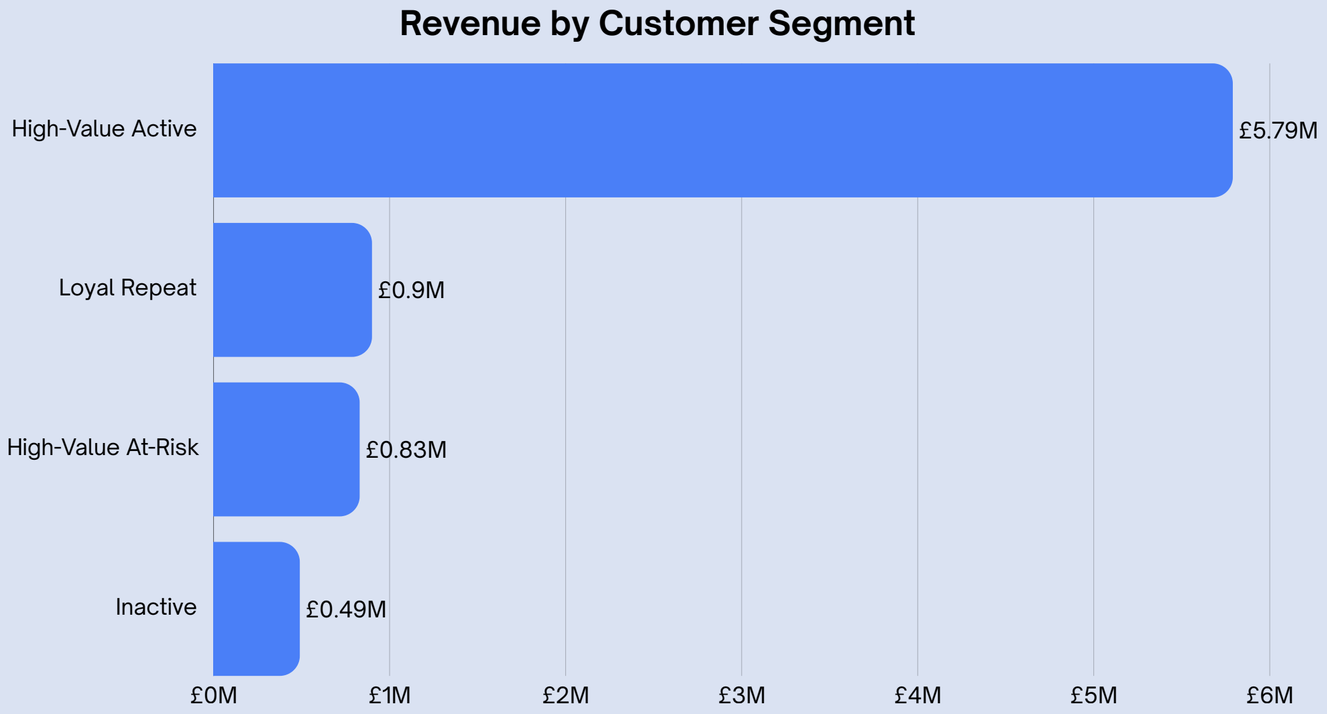
- Netherlands
- EIRE
- Germany
- France
- Australia

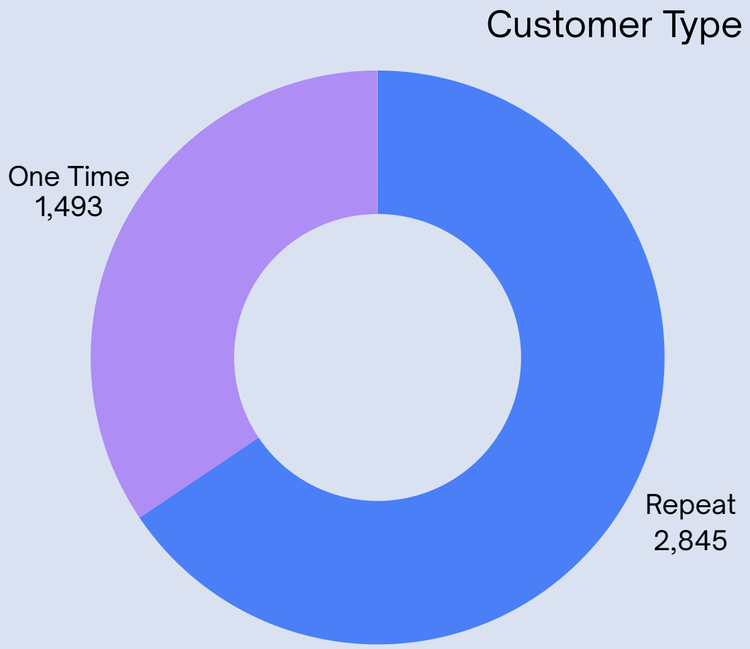
The business is highly dependent on the UK market. International markets contribute substantially less revenue individually, creating an opportunity for diversification but requiring further evaluation of customer demand, logistics, and profitability.

CUSTOMER ANALYSIS

4338	65.57%	1371	337
Customers	Repeat Customers	Inactive Customers	High-Value At-Risk

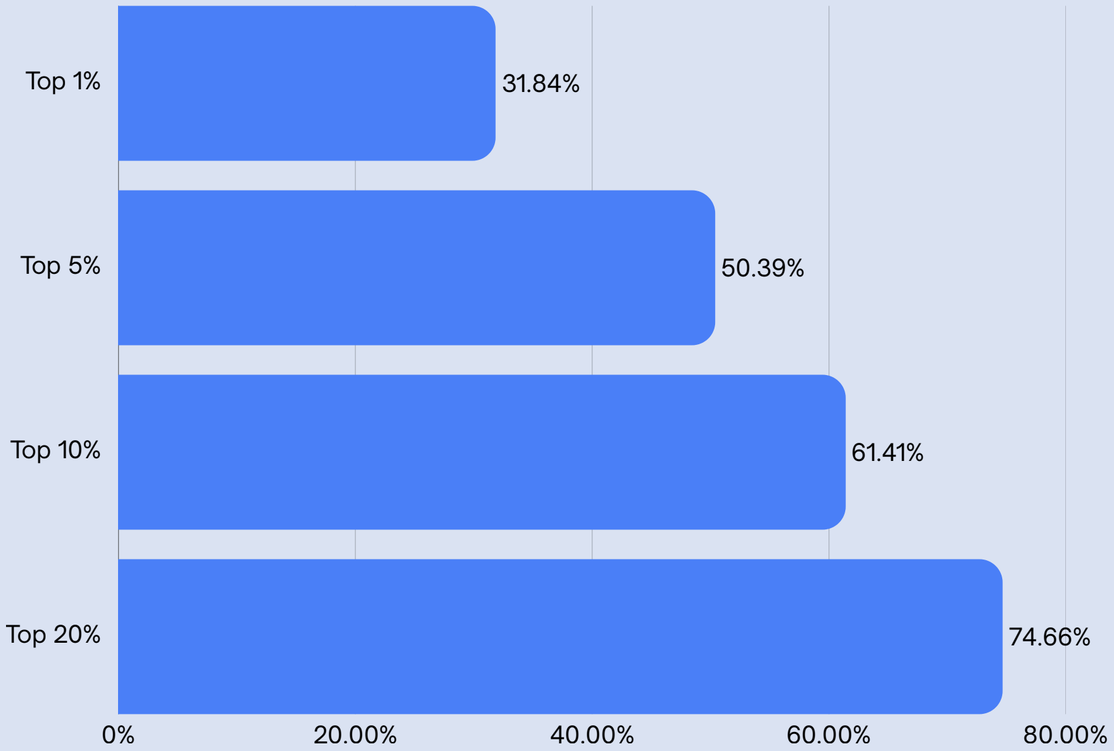
337 high-value customers are currently classified as At-Risk based on RFM behavior.





RFM analysis uses 4,338 customers because one customer had only zero-price activity and was excluded from behavioral analysis.

CUSTOMER CONCENTRATION & RISK



Revenue is meaningfully concentrated among a relatively small group of identifiable customers.

Risk signal

- 61.41% of customer-attributed revenue comes from the top 10% of customers.

This isn't automatically bad. It's a business concentration metric that management should monitor.

PRODUCT PERFORMANCE

Top merchandise products by sales revenue

Product	Revenue	Units
Regency Cakestand	£174.16K	13,862
Paper Craft	£168.47K	80,995*
White Hanging Heart	£106.42K	41,937
Party Bunting	£99.45K	18,287
Jumbo Bag Red	£94.16K	48,375

EXCEPTIONAL TRANSACTION
80,995 units in a single invoice. Validate before using for demand planning.

STRATEGIC RISKS & GROWTH LEVERS

Risk & Growth Plan

1 PROTECT THE REVENUE ENGINE

Repeat customers generate 93.09% of identifiable customer revenue, while 337 high-value customers are classified as at-risk based on RFM behavior.

2 REDUCE GEOGRAPHIC DEPENDENCY

84.59% of revenue comes from the UK.

3 SEPARATE COMMERCIAL SIGNAL FROM OPERATIONAL NOISE

Exceptional transactions and operational records can materially distort apparent sales patterns.

4 EVOLVE FROM REVENUE TO PROFITABILITY

Revenue concentration is visible, but product margin and customer profitability are not.

The next stage of growth should not simply be "sell more." It should be "protect valuable customers, diversify intelligently, improve measurement quality, and understand where profit is actually created."

Decision:

1. Build segment-based retention and reactivation programs.
2. Develop a data-driven international expansion ladder rather than broad expansion.
3. Establish separate commercial and operational KPI layers.
4. Integrate cost, margin, acquisition and retention data into future BI.

Key limitation: Profitability, product cost, acquisition cost and logistics data were unavailable; strategic recommendations therefore require validation with additional business data.

Analysis prepared using Python, SQL and Power BI

THANK YOU!

Presented By: Tamim Dihan